

Response to Reviewer

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A formal and reproducible framework for auditing observable behavior in qualitative biological network models

We thank the reviewer for a careful and constructive report. We particularly appreciate the positive assessment of the formal definitions, fixed-point algorithm, mCRL2 comparison, independent simulation game, and reproducibility materials. We agree with the central criticism: the submitted version established computational correctness more convincingly than biological utility. The revision therefore does more than add exposition. It introduces a three-interface GIM experiment on unchanged Petri-net structures, reorganizes the applications around an explicit DNA-damage question, adds verified biological support for the interface design, removes the unjustified ordinal HPN class analysis, redesigns the central figures, and separates four evidence layers throughout. Negative HPN results remain unchanged and visible.

Reviewer Comment R1

The manuscript demonstrates computational correctness much more convincingly than biological utility; what important biological conclusion can be obtained that existing comparison approaches would not reveal?

Response

We agree. The revised central result is no longer the six constructed controls. It is the interface-sensitivity analysis of the GIM animal/human and *Arabidopsis* DNA-damage models. The unchanged Petri-net structures have PN-GDDA similarity 0.9975846. Nevertheless, the behavioral result depends on the biological readout: Interfaces A and B are weakly bisimilar, while Interface C is non-comparable in both simulation directions and has $d_6 = 0.3529$. Thus a structural comparison reports nearly identical selected wiring, whereas the behavioral comparison locates the observational resolution at which the encoded correspondence breaks. This is the substantive model-level insight. We explicitly avoid extending it to organism-level equivalence.

Changes in the manuscript

The Abstract and Introduction now lead with the biological problem. Sections 4.1–4.2 formulate the question, present independent biological background, and report the structural/behavioral discrepancy. Table 3 and Fig. 3 report the executed A/B/C result. Section 5.1 interprets structural and behavioral information as complementary.

Computational/reproducibility changes

Added `src/gim_interface_analysis.py`, `results/gim_interface_sensitivity.csv`, and regression tests in `tests/test_gim_interface_analysis.py`. The result is regenerated by `make gim-interface` and `make figures`.

Status

Addressed by reframing and additional analysis

Reviewer Comment R2

The biological part should be organized around one or two explicit biological questions, moving from structural comparison to behavioral comparison, biological meaning, independent evidence, and a testable hypothesis.

Response

We agree and reorganized the applications accordingly. The primary question is: at what observational resolution do the two encoded DNA-damage control structures preserve common behavior, and when do mechanism-specific outcomes make that correspondence fail? The sequence is now biological background, structural result, coarse and intermediate behavioral results, mechanism-resolved result, interpretation, and a limited model-derived hypothesis. RCD is retained as a secondary contrasting question about directional containment, rather than being given equal evidential weight.

Changes in the manuscript

Sections 4.1–4.2 implement the requested sequence. The concluding GIM paragraph states that readouts aggregating terminal loss of proliferative capacity are expected to be more compatible with A/B, whereas readouts distinguishing apoptosis, SMR induction, and differentiation/endoreduplication should break the apparent relation. This is labeled a testable model hypothesis, not an experimental finding. Section 4.5 presents RCD as secondary.

Computational/reproducibility changes

The notebook now executes and displays the three GIM interfaces before the HPN analysis. Interface rationale, labels, internal transitions, state counts, relations, diagnostics, interpretations, and literature keys are exported to CSV.

Status

Fully addressed

Reviewer Comment R3

The GIM example is promising but needs a deeper explanation of why the comparison is biologically interesting, which functions are shared, which mechanisms differ, and what follows from weak bisimilarity.

Response

We agree. The revision distinguishes shared high-level functions from distinct implementations. Both encodings contain damage sensing, checkpoint control, repair, and recovery. The animal/human encoding includes CHK/p53-related control and apoptosis, whereas the plant encoding includes the plant-specific SOG1, WEE1 and SMR program and a combined differentiation/endoreduplication outcome. The text does not describe p53 and SOG1 as orthologs and does not equate plant programmed cell death with mammalian apoptosis. Weak bisimilarity is stated only with respect to the declared A/B observations.

Changes in the manuscript

The motivating biological background appears in the Introduction and Section 4.1. Table 2 provides process-by-process interpretation and literature support. Section 4.2 explains the matched paths and the limits of the result. The full executable nets remain available at readable scale in Appendix A (Figs. 5–6).

Computational/reproducibility changes

The plant transition previously named `CHK_signal` was renamed `SOG1_program` without changing topology or reachability. Interface C exposes separate `apoptosis`, `smr_induction`, and `differentiation_or_endoreduplication` labels.

Status

Addressed by reframing and additional analysis

Reviewer Comment R4

The observational interface is a biological modeling assumption. The manuscript should justify observable and silent events and show how conclusions change under several biologically plausible interfaces.

Response

We agree completely. Three interfaces are now defined and executed on unchanged structures. A observes coarse functions. B distinguishes DSB and replication-stress detection, ATM/ATR-associated signaling, and broad repair families while collapsing terminal fate. C additionally distinguishes the organism-specific terminal events. The outcome was not forced: A and B are weakly bisimilar; C loses weak bisimilarity and both simulations. The structural score is constant, which isolates interface resolution as the changed factor.

Changes in the manuscript

Section 4.1 defines A/B/C and Table 2 justifies each observable/internal choice. Table 3 reports both simulation directions and Fig. 3 visualizes the interface refinement. Section 5.2 states explicitly that Σ is a biological hypothesis.

Computational/reproducibility changes

Interfaces and literature keys are versioned in `src/concurrent_biomodels.py`. The files `gim_interface_sensitivity.csv`, `gim_interface_justification.csv`, and a generated LaTeX table are written deterministically. Tests assert unchanged transition sets, 13/14 states, constant PN-GDDA, and the exact A/B/C classes.

Status

Fully addressed

Reviewer Comment R5

The six construction-level cases verify implementation behavior but are not biological validation and should not be presented as such.

Response

We agree. The six cases are now consistently called a controlled construction-level software/formal benchmark. Their 6/6 result verifies that the implementation recovers the declared formal distinctions; it is not evidence of biological usefulness.

Changes in the manuscript

Section 3 and the caption of Fig. 2 state both what the benchmark establishes and what it does not. The Abstract, Introduction, Discussion, Conclusion, README, and reproducibility guide use the same terminology.

Computational/reproducibility changes

The benchmark code and expected outcomes are unchanged. Figure-generation text and documentation were revised so the output is labeled “software verification only.”

Status

Fully addressed

Reviewer Comment R6

The comparison with PN-GDDA should demonstrate complementarity rather than generic superiority, preferably on a real biological example.

Response

We agree. PN-GDDA does not attempt to represent labels, initial markings, or reachable branching, so its lower agreement on a behavioral construction task cannot establish generic inferiority. The revised GIM experiment supplies the requested biological demonstration: PN-GDDA remains 0.9976 at all interfaces, whereas the behavioral relation changes at Interface C. The methods therefore answer complementary questions.

Changes in the manuscript

The Fig. 2 caption rejects a superiority interpretation. Sections 4.2, 5.1, and the Conclusion explain the GIM structural/behavioral discrepancy and avoid “outperforms” or “better” claims.

Computational/reproducibility changes

Both 592-slot and 576-orbit PN-GDDA values remain in the machine-readable output. No threshold or score was changed to improve the result.

Status

Fully addressed

Reviewer Comment R7

The HPN-DREAM/CASPOTS analysis is too limited for strong biological conclusions and should be presented as exploratory or proof of concept; its negative results should be retained.

Response

We agree. All nine pair-condition observations and all negative results are retained. The trace association remains $\rho = -0.596$ with exact $p = 0.556$; the LTS-GDA distance association remains $\rho = 0.669$ with $p = 0.097$; and the CASPOTS native-context advantage remains unestablished at exact $p = 0.25$. We describe this analysis as an exploratory held-out compatibility check, not as validation of predictive utility, prognosis, or cell-line specificity.

Changes in the manuscript

Section 4.4 and Fig. 4 are explicitly labeled exploratory. Section 5.3 gives the sample-size and metric limitations, and the Abstract and Conclusion preserve the negative interpretation.

Computational/reproducibility changes

No favorable alternative test was sought. The existing exact permutations for the two predeclared continuous diagnostics and the exact CASPOTS sign test were retained.

Status

Addressed within the limitations of the available data

Reviewer Comment R8

The formal relations do not form an obvious one-dimensional ordinal scale, especially because one-way simulation is directional. The Spearman encoding risks discarding the information that motivates simulation.

Response

We agree and removed the ordinal analysis rather than defending it. Strong/weak equivalence, directional simulation, and non-comparability do not have a biologically justified total order. The revised output preserves three left-in-right relations, three right-in-left relations, and three pairs with no simulation. Because the experimental pair RMSE is symmetric and $n = 9$, we make no direction-aware inferential claim. Descriptive medians are shown without a scalar class test.

Changes in the manuscript

The former $\rho = 0.091$, $p = 0.111$ result is removed from the manuscript and all public documentation because the analysis itself was invalid, not because it was negative. Section 4.4 explains the decision; Fig. 4b displays the nominal directions; Section 5.3 discusses the limitation.

Computational/reproducibility changes

`src/hpn_dream_validation.py` no longer creates ordinal class scores or formal-class Spearman fields. It writes `relation_direction` and nominal summaries. Regression tests require 3/3/3 directions and explicitly reject the obsolete fields. The result JSON records `formal_class_scalar_encoding_used=false` and `formal_class_inferential_test_performed=false`.

Status

Fully addressed

Reviewer Comment R9

Machine-checked transition provenance establishes reproducibility of modeling choices, not that the resulting curated Petri nets are biologically correct.

Response

We agree. The revision distinguishes provenance, internal reproducibility, executable-model validation, and external validation of a biological interpretation. The curated models are described as literature-informed encodings without kinetics, concentrations, tissue context, stochasticity, or independent experimental calibration.

Changes in the manuscript

Section 4.5 states what transition-level provenance does and does not establish. Section 5.3 and Table 7 separate the four evidence layers. GIM conclusions are labeled as model-derived, literature-consistent, or testable rather than experimentally validated.

Computational/reproducibility changes

Provenance coverage tests are retained, but documentation no longer promotes coverage to biological validation. The GIM justification CSV separately records biological rationale and literature keys.

Status

Fully addressed

Reviewer Comment R10

The benchmark and central biological figures are difficult to read at manuscript scale and require conceptual redesign rather than a higher resolution export.

Response

We agree. Both figures were regenerated as vector artwork with larger type and less visual density. The benchmark is split into a decision matrix and a simple agreement panel, explicitly labeled as software verification. The central GIM figure no longer asks readers to decode two compressed Petri nets: it shows shared functions, organism-specific implementations, matched coarse control flow, and the A/B/C classification. Full nets are retained in the appendix at readable width.

Changes in the manuscript

Revised Figs. 2–3 and their captions replace the submitted figures. Fig. 4 was also redesigned to preserve HPN simulation direction instead of displaying an ordinal class plot. Appendix Figs. 5–6

retain the detailed Petri nets.

Computational/reproducibility changes

All artwork is generated by `make_figures.py`; the main scientific figures are PDF vectors with PNG previews for inspection.

Status

Fully addressed

Reviewer Comment R11

The manuscript should separate formal correctness, software implementation validation, validation of executable biological models, and biological validation of conclusions.

Response

We agree and adopted this four-layer taxonomy throughout. Formal proofs belong to Layer I; construction controls and independent software oracles to Layer II; hashes, ingestion controls, provenance, and semantics to Layer III; and independent biological literature or held-out observations to Layer IV. Each layer now has an explicit boundary.

Changes in the manuscript

The taxonomy is introduced in the Introduction and formalized in Discussion Table 7. The Abstract, Section 3, Section 4, Discussion, and Conclusion use the same distinctions.

Computational/reproducibility changes

README, REPRODUCIBILITY.md, and the executed notebook now reproduce the same evidence terminology and identify which artifact supports each layer.

Status

Fully addressed

Reviewer Comment R12

The manuscript currently reads as a formal-methods framework illustrated with biological models; the revision should make one or two carefully supported biological case studies stand out without weakening the formal contribution.

Response

We agree with this overall assessment. The formal framework, proofs, simulation orientation, and external oracles are preserved. The narrative now begins with the biological model-comparison problem, uses GIM as the central case, treats RCD as a limited directional contrast, demotes transformed public models to ingestion controls, and presents HPN as exploratory evidence. The conclusion is now a biologically meaningful but restrained statement about locating the observational boundary between high-level encoded function and mechanism-specific behavior.

Changes in the manuscript

The Abstract, Introduction, all of Section 4, Discussion, and Conclusion were rewritten. Supporting examples are compact, GIM is developed in depth, and all organism-level or predictive overinterpretations are expressly excluded.

Computational/reproducibility changes

The revised manuscript, notebook, figures, deterministic result tables, tests, and submission package are generated from the same repository state. mCRL2 202607.0 was rerun on the public, synthetic, and HPN LTSs.

Status

Addressed by reframing and additional analysis

Summary of verification performed

The complete notebook executed successfully. All 46 regression tests passed with mCRL2 available. The revised JBCB source compiles without undefined citations or references. The final source package is also compiled independently from a flat directory, matching the journal upload constraint.